6.3 Sine, Cosine, and Tangent (SOH-CAH-TOA)

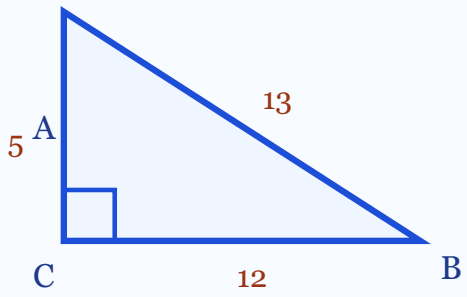
For an acute angle θ in a right triangle, label the sides relative to θ: the **hypotenuse** (always opposite the right angle), the **opposite** side (across from θ), and the **adjacent** side (next to θ, not the hypotenuse). The three basic trig ratios are $\sin\theta = \dfrac{\text{opp}}{\text{hyp}}$, $\cos\theta = \dfrac{\text{adj}}{\text{hyp}}$, $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$ — remembered by **SOH-CAH-TOA**. Because all right triangles with the same acute angle are similar, these ratios depend only on the angle, never on the triangle's size. A useful consequence: the two acute angles A and B of a right triangle are complementary ($A + B = 90°$), and the side opposite A is the same side that's adjacent to B — so $\sin A = \cos B$ always. This worksheet practises writing ratios, solving for missing sides, and combining trig with the Pythagorean Theorem.

Examples

Example 1: Write $\sin$

Right $\triangle ABC$ has a right angle at C. The side opposite $\angle A$ is 5, the side adjacent to $\angle A$ is 12, and the hypotenuse is 13. Find $\sin A$.

Solution



Relative to $\angle A$: opposite $= 5$, adjacent $= 12$, hypotenuse $= 13$.

Identify the parts relative to $\angle A$: opposite $= 5$, hypotenuse $= 13$.

Apply SOH: $\sin A = \dfrac{\text{opp}}{\text{hyp}}$

$\sin A = \dfrac{5}{13}$.

Answer: $\sin A = \dfrac{5}{13}$.

Example 2: Write $\cos$

Using the same triangle as Example 1 (opposite $= 5$, adjacent $= 12$, hypotenuse $= 13$), find $\cos A$.

Solution

Identify the parts relative to $\angle A$: adjacent $= 12$, hypotenuse $= 13$.

Apply CAH: $\cos A = \dfrac{\text{adj}}{\text{hyp}}$

$\cos A = \dfrac{12}{13}$.

Answer: $\cos A = \dfrac{12}{13}$.

Example 3: Write $\tan$

A right triangle has, relative to angle θ: opposite $= 8$, adjacent $= 15$. Find $\tan\theta$.

Solution

Identify the parts: opposite $= 8$, adjacent $= 15$.

Apply TOA: $\tan\theta = \dfrac{\text{opp}}{\text{adj}}$

$\tan\theta = \dfrac{8}{15}$.

Answer: $\tan\theta = \dfrac{8}{15}$.

Example 4: Find the opposite side

$\sin\theta = \dfrac{3}{5}$ and the hypotenuse is 20. Find the opposite side.

Solution

Set up the ratio: $\sin\theta = \dfrac{\text{opp}}{\text{hyp}} \Rightarrow \dfrac{3}{5} = \dfrac{\text{opp}}{20}$

Solve: $\text{opp} = 20 \times \dfrac{3}{5} = 12$.

Answer: 12.

NAME: _____ Geometry (FL B.E.S.T.)

Example 5: Find the adjacent side

$\cos\theta = \frac{4}{5}$ and the hypotenuse is 15. Find the adjacent side.

Solution

Set up the ratio: $\cos\theta = \frac{\text{adj}}{\text{hyp}} \Rightarrow \frac{4}{5} = \frac{\text{adj}}{15}$

Solve: $\text{adj} = 15 \times \frac{4}{5} = 12$.

Answer: 12.

Example 6: Find the opposite side using $\tan$

$\tan A = \frac{3}{4}$ and the adjacent side is 8. Find the opposite side.

Solution

Set up the ratio: $\tan A = \frac{\text{opp}}{\text{adj}} \Rightarrow \frac{3}{4} = \frac{\text{opp}}{8}$

Solve: $\text{opp} = 8 \times \frac{3}{4} = 6$.

Answer: 6.

Example 7: Why $\sin A=\cos B$

In a right triangle with right angle C, prove that $\sin A = \cos B$ where A and B are the two acute angles.

Solution

Since $\angle C = 90°$, the Angle-Sum Theorem gives $\angle A + \angle B = 90°$ — the acute angles are complementary.

The side opposite $\angle A$ (call it a) is the same side that touches $\angle B$ but is not the hypotenuse — so a is *adjacent* to $\angle B$.

Therefore $\sin A = \frac{a}{\text{hyp}}$ and $\cos B = \frac{a}{\text{hyp}}$ — the same ratio.

Numeric check with $a = 5$, $\text{hyp} = 13$: $\sin A = \frac{5}{13}$ and $\cos B = \frac{5}{13}$ ✓.

Conclusion: $\sin A = \cos B$ always, for complementary acute angles.

Example 8: Find the hypotenuse

$\sin\theta = \frac{5}{13}$ and the opposite side is 10. Find the hypotenuse.

Solution

Set up the ratio: $\sin\theta = \frac{\text{opp}}{\text{hyp}} \Rightarrow \frac{5}{13} = \frac{10}{\text{hyp}}$

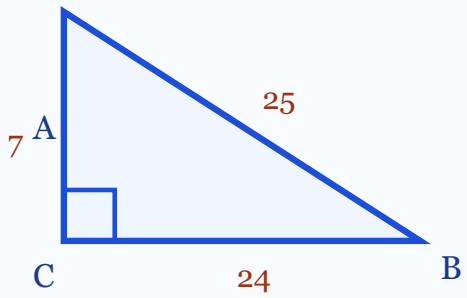
Solve: $\text{hyp} = 10 \div \frac{5}{13} = 10 \times \frac{13}{5} = 26.$

Answer: 26.

Example 9: All three ratios from a 7-24-25 triangle

A right triangle has opposite $= 7$, adjacent $= 24$, hypotenuse $= 25$ (relative to θ). Find $\sin\theta$, $\cos\theta$, and $\tan\theta$.

Solution



A 7-24-25 right triangle.

Apply each ratio directly from the given sides:

$\sin\theta = \frac{7}{25}$

$\cos\theta = \frac{24}{25}$

$\tan\theta = \frac{7}{24}.$

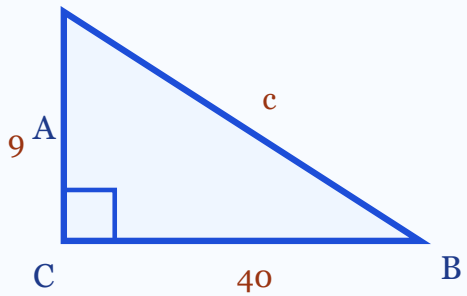
Answer: $\sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}, \tan\theta = \frac{7}{24}.$

NAME: _____ Geometry (FL B.E.S.T.)

Example 10: Pythagorean Theorem, then trig

A right triangle has legs 9 and 40. Find the hypotenuse, then $\sin\theta$ and $\cos\theta$ for the acute angle opposite the leg of length 9.

Solution



First find c, then use all three sides for the trig ratios.

Find the hypotenuse using the Pythagorean Theorem: $c^2 = 9^2 + 40^2 = 81 + 1600 = 1681$

$c = \sqrt{1681} = 41$.

Now apply the trig ratios relative to the angle opposite 9: $\sin\theta = \dfrac{9}{41}$; $\cos\theta = \dfrac{40}{41}$.

Answer: hypotenuse $= 41$; $\sin\theta = \dfrac{9}{41}$, $\cos\theta = \dfrac{40}{41}$.

NAME: _____ Geometry (FL B.E.S.T.)

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: $\sin$

Opposite $= 9$, hypotenuse $= 15$. Find $\sin\theta$.

Answer: **Verification:**

Problem 2: $\cos$

Adjacent $= 8$, hypotenuse $= 17$. Find $\cos\theta$.

Answer: **Verification:**

Problem 3: $\tan$

Opposite $= 5$, adjacent $= 12$. Find $\tan\theta$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 4: Find opposite

$\sin\theta = \frac{4}{5}$, hypotenuse $= 25$. Find the opposite side.

Answer: **Verification:**

Problem 5: Find adjacent

$\cos\theta = \frac{3}{5}$, hypotenuse $= 20$. Find the adjacent side.

Answer: **Verification:**

Problem 6: Find opposite

$\tan\theta = \frac{5}{12}$, adjacent $= 24$. Find the opposite side.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 7: Find hypotenuse

$\sin\theta = \frac{8}{17}$, opposite $= 16$. Find the hypotenuse.

Answer: **Verification:**

Problem 8: All ratios

Opposite $= 6$, adjacent $= 8$, hypotenuse $= 10$. Find $\sin\theta$, $\cos\theta$, $\tan\theta$.

Answer: **Verification:**

Problem 9: Find adjacent

$\tan\theta = \frac{7}{24}$, opposite $= 14$. Find the adjacent side.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

Problem 10: $\cos$

Adjacent $= 20$, hypotenuse $= 25$. Find $\cos\theta$.

Answer: **Verification:**

Problem 11: Find hypotenuse

$\cos\theta = \frac{12}{13}$, adjacent $= 36$. Find the hypotenuse.

Answer: **Verification:**

Problem 12: Complementary

If $\sin A = \frac{5}{13}$, what is $\cos B$ for the other acute angle B?

Answer: **Verification:**

Problem 13: Challenge — Pythagorean + trig

A right triangle has legs 20 and 21. Find the hypotenuse, then $\sin\theta$ and $\cos\theta$ for the angle opposite the leg of 20.

Answer: **Verification:**

Problem 14: Challenge — from $\tan$ to all ratios

$\tan\theta = \frac{3}{4}$. Find $\sin\theta$ and $\cos\theta$.

Answer: **Verification:**

Problem 15: Challenge — multi-step

A right triangle has hypotenuse 50 and $\sin\theta = 0.6$ for one acute angle. Find the opposite side, then the adjacent side (Pythagorean Theorem), then $\cos\theta$.

Answer: **Verification:**

NAME: _____ Geometry (FL B.E.S.T.)

✓ Answer Key

1. $\frac{3}{5}$
2. $\frac{8}{17}$
3. $\frac{5}{12}$
4. 20
5. 12
6. 10
7. 34
8. $\frac{3}{5}$; $\frac{4}{5}$; $\frac{3}{4}$
9. 48
10. $\frac{4}{5}$
11. 39
12. $\frac{5}{13}$
13. Hypotenuse = 29; $\sin\theta = \frac{20}{29}$, $\cos\theta = \frac{21}{29}$
14. $\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$
15. Opposite = 30; adjacent = 40; $\cos\theta = \frac{4}{5}$

Great work — keep practising!